

EXPERIMENT – 5

Title: To Study Different Constraints

Aim / Objective: To understand and implement various SQL constraints such as **PRIMARY KEY, FOREIGN KEY, UNIQUE, CHECK, NOT NULL, and DEFAULT** to maintain data integrity and consistency in a relational database.

Theory: Constraints are rules applied to table columns that enforce the accuracy, validity, and integrity of data stored in a database. They prevent invalid data entry and maintain relationships between tables.

Constraints can be defined either during table creation using the **CREATE TABLE** statement or later using the **ALTER TABLE** statement.

The commonly used constraints are:

- PRIMARY KEY
- FOREIGN KEY
- UNIQUE
- CHECK
- NOT NULL
- DEFAULT

Types of Constraints

1. PRIMARY KEY: A Primary Key uniquely identifies each record in a table.

Characteristics

- Cannot contain NULL values.
- Must contain unique values.
- Only one Primary Key is allowed per table.

2. FOREIGN KEY: A Foreign Key establishes a relationship between two tables by referring to the Primary Key of another table.

Characteristics

- Maintains referential integrity.
- Prevents invalid references.
- Can contain duplicate values.

3. UNIQUE Constraint: Ensures that all values in a column are unique.

Characteristics

- Prevents duplicate values.
- Allows NULL values (depending on the DBMS).

4. CHECK Constraint: Ensures that values satisfy a specified condition.

Example

- Age ≥ 18
- Salary > 10000

5. NOT NULL Constraint: Ensures that a column cannot contain NULL values.

6. DEFAULT Constraint: Assigns a default value if no value is provided during insertion.

Learning Objectives: After completing this experiment, students will be able to:

- Understand different SQL constraints.
- Apply constraints while creating tables.
- Maintain data integrity.
- Prevent invalid data entry.
- Establish relationships between tables.

Software Requirements

- Oracle Database
- Oracle SQL Developer / SQL*Plus

Algorithm / Procedure

1. Open Oracle SQL Developer.
2. Connect to the Oracle database.
3. Create Department table.
4. Create Employee table with constraints.
5. Insert records.
6. Verify each constraint.
7. Observe constraint violations.
8. Stop.

Program

Step 1 : Create Department Table

```
CREATE TABLE Department( DeptID NUMBER PRIMARY KEY, DeptName  
VARCHAR2(30) UNIQUE );
```

Step 2 : Insert Department Records

```
INSERT INTO Department VALUES(101,'IT');  
INSERT INTO Department VALUES(102,'HR');  
INSERT INTO Department VALUES(103,'Finance');  
COMMIT;
```

Step 3 : Create Employee Table

```
CREATE TABLE Employee ( EmpID NUMBER PRIMARY KEY, EmpName  
VARCHAR2(30) NOT NULL, Email VARCHAR2(50) UNIQUE, Age NUMBER  
CHECK(Age>=18), Salary NUMBER CHECK(Salary>=10000), Department  
VARCHAR2(20) DEFAULT 'General', DeptID NUMBER, CONSTRAINT FK_Department  
FOREIGN KEY(DeptID) REFERENCES Department(DeptID) );
```

Display Table Structure: DESC Employee;

Insert Valid Records: INSERT INTO Employee VALUES

```
(101,'Rahul','rahul@gmail.com',24,35000,'IT',101);  
INSERT INTO Employee VALUES (102,'Priya','priya@gmail.com',23,42000,'HR',102);  
COMMIT;
```

Verify PRIMARY KEY Constraint

```
INSERT INTO Employee VALUES  
(101,'Aman','aman@gmail.com',24,30000,'Finance',103);
```

Expected Output: ORA-00001: unique constraint violated

Verify UNIQUE Constraint

INSERT INTO Employee VALUES
(103,'Karan','rahul@gmail.com',24,35000,'IT',101);

Expected Output: ORA-00001: unique constraint violated

Verify NOT NULL Constraint

INSERT INTO Employee VALUES
(104,NULL,'abc@gmail.com',24,30000,'HR',102);

Expected Output: ORA-01400: cannot insert NULL

Verify CHECK Constraint

INSERT INTO Employee VALUES
(105,'Riya','riya@gmail.com',16,25000,'IT',101);

Expected Output: ORA-02290: check constraint violated

Verify Salary CHECK Constraint

INSERT INTO Employee VALUES
(106,'Neha','neha@gmail.com',25,8000,'Finance',103);

Expected Output: ORA-02290: check constraint violated

Verify FOREIGN KEY Constraint

INSERT INTO Employee VALUES
(107,'Mohit','mohit@gmail.com',24,35000,'Sales',110);

Expected Output: ORA-02291: integrity constraint violated

Verify DEFAULT Constraint

INSERT INTO Employee (EmpID, EmpName, Email, Age, Salary, DeptID) VALUES
(108,'Deepak','deepak@gmail.com',25,40000,101);
COMMIT;

Display Records

SELECT * FROM Employee;

Expected Output

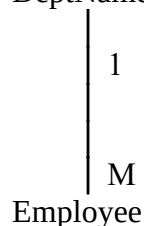
EmpID	EmpName	Email	Age	Salary	Department	DeptID
101	Rahul	rahul@gmail.com	24	35000	IT	101
102	Priya	priya@gmail.com	23	42000	HR	102
108	Deepak	deepak@gmail.com	25	40000	General	101

Entity Relationship Diagram

Department

DeptID (PK)

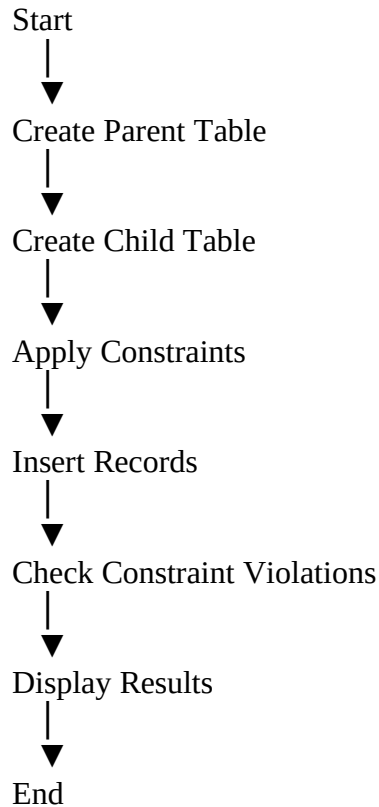
DeptName



Employee

EmpID (PK) ,EmpName (NOT NULL), Email (UNIQUE), Age (CHECK), Salary (CHECK),
Department (DEFAULT), DeptID (FK)

Flowchart



Explanation:

PRIMARY KEY: Uniquely identifies every employee.

FOREIGN KEY: Links Employee table with Department table.

UNIQUE: Prevents duplicate email addresses.

CHECK: Validates Age and Salary values.

NOT NULL: Ensures employee name is mandatory.

DEFAULT: Assigns the value **General** if Department is not specified.

Observation

- Primary Key prevented duplicate employee IDs.
- Foreign Key maintained referential integrity.
- UNIQUE prevented duplicate email addresses.
- CHECK validated age and salary values.
- NOT NULL prevented null employee names.
- DEFAULT automatically assigned a department when omitted.

Result:

Various SQL constraints including **PRIMARY KEY, FOREIGN KEY, UNIQUE, CHECK, NOT NULL, and DEFAULT** were successfully implemented. These constraints ensured data accuracy, consistency, and integrity within the database.

Applications

- Banking Systems
- Hospital Management Systems
- Payroll Systems
- Student Information Systems
- Inventory Management Systems
- Railway Reservation Systems
- E-Commerce Applications

Advantages

- Ensures data integrity.
- Prevents duplicate records.
- Maintains referential integrity.
- Improves database reliability.
- Reduces data inconsistency.
- Enforces business rules.

Viva Questions

1. What is a constraint in SQL?
2. What is the purpose of a Primary Key?
3. What is the difference between PRIMARY KEY and UNIQUE?
4. What is a Foreign Key?
5. What is referential integrity?
6. What is the CHECK constraint?
7. What is the NOT NULL constraint?
8. What is the DEFAULT constraint?
9. Can a table have more than one UNIQUE constraint?
10. Why are constraints important in a relational database?

Lab Assignment

1. Create a **Student** table using all constraints.
2. Create **Customer** and **Order** tables with a Foreign Key relationship.
3. Apply a CHECK constraint to validate marks greater than or equal to 0.
4. Apply a UNIQUE constraint on the mobile number.
5. Apply a DEFAULT value for the city column.
6. Insert valid and invalid records to observe constraint behavior.
7. Compare PRIMARY KEY and FOREIGN KEY.
8. Explain the role of each constraint with examples.
9. Create an **Employee_Project** table with appropriate constraints.
10. Write SQL queries to verify each constraint.

Precautions

- Create the parent table before creating the child table.
- Insert parent table records before inserting child table records.
- Ensure Primary Key values are unique.

- Verify CHECK conditions before inserting data.
- Use meaningful names for constraints.
- Test each constraint individually to understand its behavior.

Learning Outcome

After completing this experiment, students will be able to:

- Apply different SQL constraints while designing database tables.
- Maintain data integrity using PRIMARY KEY, FOREIGN KEY, UNIQUE, CHECK, NOT NULL, and DEFAULT constraints.
- Establish relationships between tables.
- Prevent invalid and duplicate data entries.
- Design reliable and consistent relational databases for real-world applications.